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Z. Y. Cheng; C. S. Tsai

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Optically activated integrated optic Mach–Zehnder interferometer on GaAs

Z. Y. Cheng and C. S. Tsai
Department of Electrical and Computer Engineering and Institute for Surface and Interface Science, University of California, Irvine, California 92717

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An optically activated guided-wave Mach–Zehnder interferometer has been constructed and used to study photoinduced phase modulation in GaAs channel waveguides. A π radian phase shift at the optical wavelength of 1.15 μm has been achieved with modulating pulse energy of 320 pJ at an interaction length of 22 μm. Our experiment also indicates that for infrared (IR) light with photon energy far below the band-gap energy of the semiconductor material, the free carrier-induced refractive index changes are much more significant than the free carrier-induced absorption changes.

Photoconductive property of semiconductors has been widely used to achieve optically controlled subpicosecond switching and wideband modulation in microwave devices.[1] In such applications, the optically generated free carriers cause significant changes in the conductivity of the semiconductor material and, thus, facilitate efficient control of the propagation of the microwaves in the devices. Significant refractive index changes in the optical region caused by photoinduced free carriers in semiconductors have also been observed and measured.[2,3] It is, therefore, possible to construct optical-to-optical modulation and switching devices using the principle similar to that employed for the microwave devices. A couple of reports on all-optical switching and modulation that utilized photoinduced changes in refractive index and photoinduced changes in absorption in channel- and planar-waveguides on silicon substrate[4,5] have appeared recently. Similar experiments on GaAs waveguides that involve photoinduced absorption changes have also been reported most recently.[6] Earlier we had also carried out optically activated cutoff modulation and switching in GaAs channel waveguides.[7]

The capability of inducing large refractive index changes in III-V semiconductor compounds by photoinduced free carriers enables construction of efficient optically controlled optical phase modulators. In this letter, we report on an optical-to-optical Mach–Zehnder interferometric modulator on a GaAs substrate which derives the required phase shift via optical activation. Figure 1 shows the device structure. The single-mode Mach–Zehnder interferometer was fabricated on a double-epilayer structure composed of a 1.0 μm undoped GaAs layer and a 3.0 μm $Al_{0.05}Ga_{0.95}As$ layer grown on an n^+-GaAs substrate. The lateral optical confinement was achieved by using a ridge waveguide structure with a channel width of 5.0 μm and a ridge height of 0.4 μm which creates a lateral effective index step of about 0.0045 (see inset in Fig. 1). The interferometer was found to support single TE- and TM-mode propagation at the wavelength of 1.15 μm, and at the TE-mode propagation, the excess loss of the interferometer over the straight channel waveguide of identical length fabricated on the same wavelength substrate was measured to be 2.0 dB.

In the experiment, a laser beam at the wavelength of 1.15 μm was edge-coupled into the interferometer. A modulating light with photon energy higher than the band-gap energy of GaAs was focused upon one of the arms of the interferometer as shown in Fig. 1. Two modulating light sources were used in the experiment: first, a cw 0.6328 μm He-Ne laser and then a pulsed excimer laser-pumped dye laser with the wavelength of 550 nm. The focused spot size of the modulating He-Ne laser light upon the sample was 22 μm. Absorption of the modulating light in the GaAs channel waveguide resulted in generation of free carriers via direct interband absorption. The excess free carriers in turn cause changes in the refractive index of the waveguide through the plasma effect and the band filling (or band-gap shift) effect and, therefore, result in a phase shift in the guided-light propagating in the illuminated arm of the interferometer. The induced phase shift was then converted into intensity modulation in the output of the interferometer. Figure 2 shows a typical output of the optically activated interferometer with the cw 0.6328 μm He-Ne modulating light mechanically chopped at 1.3 KHz. The possible effect of a photoinduced free-carrier absorption was also examined by illuminating the same modulating light upon a series of straight channel waveguide of identical length and width fabricated on the same substrate. However, since the interaction length was only 22 μm and the modulating light power was rather low, the effect of free-carrier absorption was too weak to detect. Therefore, from the measured intensity changes in the output of the

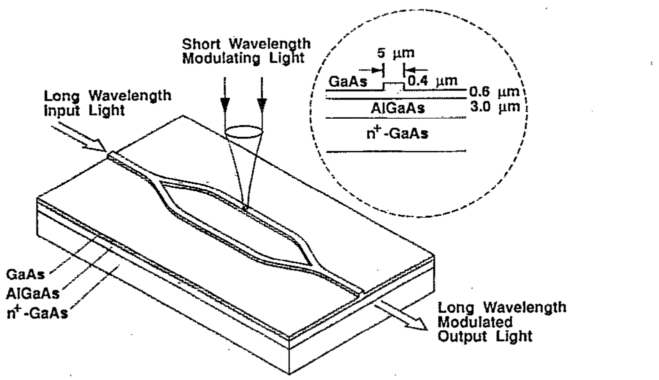

FIG. 1. Structure of optically activated Mach–Zehnder interferometric modulator in GaAs.

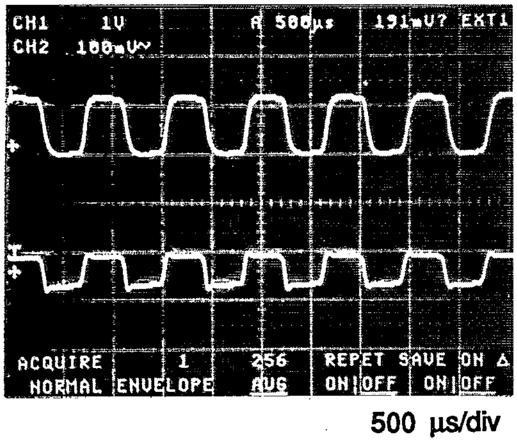

FIG. 2. Typical waveform of the output signal of the optically activated interferometer. Upper: modulated light waveform; bottom: modulating light waveform.

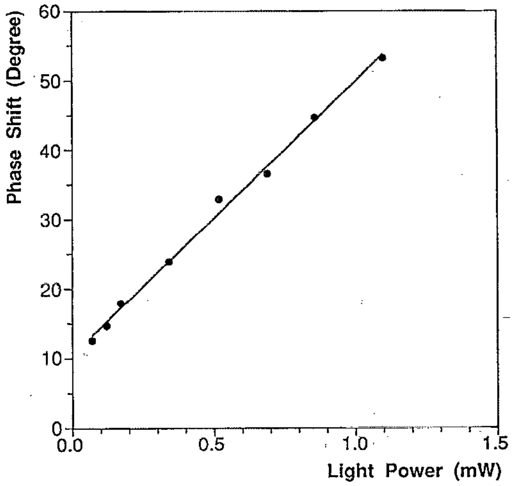

FIG. 4. Photoinduced phase shift in the guided-light vs modulating light power with interaction length of 22 μm.

interferometer the corresponding photoinduced phase shift can be deduced. Figure 3 shows the measured modulation depth of the output signal from the interferometer versus the effective modulating light power. The effective modulating power P_{eff} was estimated by

$$P_{\text{eff}}=\frac{A_{\text{ch}}(1-R)}{A_{\text{sp}}}P_{\text{in}}=\xi P_{\text{in}},$$

where P_{in} is the incident power of the modulating light; A_{ch} is the area of the portion of the channel waveguide illuminated by the modulating light, and A_{sp} is the area of the modulating light spot focused on the sample; and R is the reflection coefficient of GaAs at the wavelength of 0.6328 μm. In the experiment, the maximum cw modulating light power incident upon the channel waveguide was about 1.1 mW. The resulting modulation depth was 20% which corresponds to a phase shift of 53° in the interaction length of 22 μm. Figure 4 shows the corresponding phase shift as a function of the modulating light power, clearly indicating a linear dependence of the photoinduced phase shift on the effective modulating light power. This observation is in agreement with the linear dependence of the refractive index changes on the photoinduced free-carrier concentration which has been reported by a number of researchers.[8–10]

In order to further increase the concentration of the photoinduced free carriers and thus the photoinduced phase shift, a pulsed dye laser (at a wavelength of 550 nm and pulse width of 30 ns) of considerably higher energy was used as the modulating light in the experiment. We measured the photoinduced phase shift versus the energy of the modulating laser pulse as it determines the total number of photons involved in the interaction.[4,6] Figure 5 shows the measured modulation depth as a function of the modulating pulse energy. The maximum modulation depth measured in the interferometer output was about 85% at a modulating pulse energy of 320 pJ. In the measurement, we carefully identified the modulating pulse energy level at

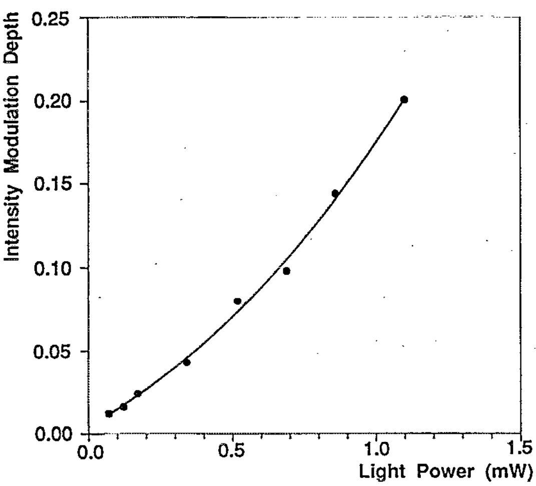

FIG. 3. Modulation depth of the output signal of the optically activated Mach–Zehnder interferometer vs modulating light power with interaction length of 22 μm.

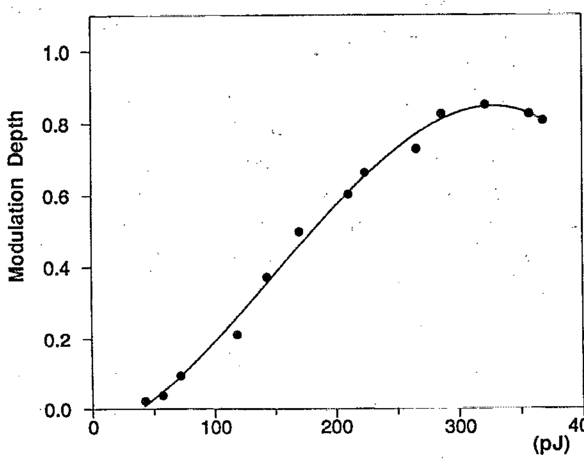

FIG. 5. Modulation depth of the output signal of the optically activated interferometer vs the modulating light pulse energy.

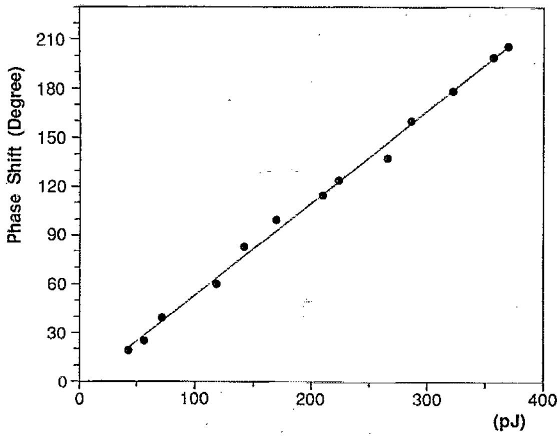

FIG. 6. Photoinduced phase shift in the guided-light vs modulating light pulse energy with an interaction length of 22 μm.

which the maximum modulation depth was achieved. We noticed that if the pulse energy was increased beyond that level the output of the interferometer began to decrease as shown in Fig. 5. This observation indicates that at this modulating energy level the photoinduced phase shift has reached π radians and the light beams guided in the two arms of the interferometer were out of phase. This result should be compared with that reported in Ref. 6, where the all-optical modulation experiment was carried out using straight channel waveguides on GaAs and the modulation was due to the free carrier-induced absorption changes rather than refractive index changes. Since the photon energy of the guided IR light (at 1.15 μm wavelength) is far below the band-gap energy of the material, free carrier-induced refractive index changes should dominate over the absorption changes.[3] In the experiment reported in Ref. 6, a 90% modulation depth was measured at modulating pulse energy of 30 nJ which is almost two-orders of magnitude higher than that used to obtain a π-phase shift in our experiment. Although the wavelengths of the IR guided light used in the two experiments are slightly different (1.3 and 1.15 μm), a comparison of the two results indicates that the effect of photoinduced refractive index changes (or phase shift) rather than the absorption changes was dominant in our experiment.

The corresponding photoinduced phase shift deduced from the measured data is shown in Fig. 6, clearly indicating a linear dependence of the photoinduced phase shift on the effective modulating light energy in our experiment. Again, this observation was in agreement with the linear dependence of the refractive index changes on the photoinduced free-carrier concentration which has been reported by a number of researchers.[8–10]

Similar to their microwave counterparts, the ultimate limitation on the speed of these types of optical devices is the free-carrier lifetime of the semiconductor material involved. For undoped GaAs the carrier lifetime is about 10 ns (Ref. 11) and the speed of the resulting devices is limited to a similar range. However, the carrier lifetime can be substantially reduced by doping or ion implantation of recombination centers into the material. Carrier lifetimes in the range of picoseconds to subpicoseconds have been achieved in GaAs by using doping or ion implantation techniques.[1,12] Doping and ion implantation will increase the attenuation and/or cause damage to the optical waveguide. Therefore, a tradeoff between the loss and the speed of the devices must be made. As mentioned above, the interaction length of this type of device can be very short. We may dope or implant only locally along the channel waveguide. By carefully controlling the region to be processed and properly choosing the dose level, very efficient and wideband optical-to-optical modulators and switches can be realized.

In summary, an optically activated Mach–Zehnder interferometer has been constructed and used to study photoinduced phase modulation in GaAs channel waveguides. A π radian photoinduced phase shift has been achieved with a modulating pulse energy of 320 pJ at an interaction length of 22 μm. Our study also shows that for an IR light with photon energy far below the band-gap energy of the semiconductor material, the free carrier-induced refractive index change is much more significant than the free carrier-induced absorption changes.

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